COURSE: DSA 2020

COURSE TITLE: INTRODUCTION TO ARTIFICIAL INTELLIGENCE

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Document: AI Term Projects

These projects are designed to help you apply your knowledge and skills in AI, focusing on basic machine learning, simple algorithms, and real-world applications in Africa. You will learn how AI can solve practical problems while considering ethical and technical challenges. The students can pair to do each project.

Project 1: AI-Powered Chatbot for Local Businesses

Objective:

To develop a simple AI chatbot to assist small African businesses in answering frequently asked customer questions.

Tasks:

1. Collect common customer questions from a specific business sector (e.g., retail, agriculture, or healthcare).
2. Use rule-based or basic NLP techniques to develop a chatbot.
3. Train the chatbot using a small dataset of common customer inquiries and responses.
4. Test the chatbot and improve its responses based on user feedback.
5. Discuss challenges in training AI with limited local language data.

Expected Outcome:

- A working chatbot that can answer basic customer questions.
- A demonstration of the chatbot in a real-world scenario.
- A short report on challenges and potential improvements.

Project 2: Intelligent Path Finder for Nairobi Matatu Routes

Objective: Apply search algorithms to solve a real-world Kenyan transportation problem, demonstrating uninformed and informed search efficiency while addressing urban mobility challenges in Nairobi.

Key Tasks:

- Model a simplified Nairobi matatu network (10–20 nodes/stops, e.g., CBD → Westlands → Rongai) as a graph with edge costs (distance/time).
- Implement BFS, DFS, and A* (with at least one heuristic like straight-line distance).
- Compare algorithms on path length, nodes expanded, and runtime.
- Visualize routes (text or simple matplotlib plot).

Expected Outcomes:

- Working code showing optimal routes.
- Comparison table/graphs.
- Report discussing algorithm suitability for Kenyan traffic variability.
- Demo of finding routes with explanations.

Project 3: AI Player for Tic-Tac-Toe or Simplified Bao Game

Objective: Implement adversarial search to create an intelligent game-playing agent, exploring minimax in a culturally relevant context (Bao as a traditional East African game).

Key Tasks:

- Build game logic for Tic-Tac-Toe or simplified Bao (e.g., smaller board).
- Implement minimax algorithm (± alpha-beta pruning optional).
- Add difficulty levels by depth limit.
- Enable human vs AI play.

Expected Outcomes:

- Unbeatable/optimal AI play demonstration.
- Explanation of minimax tree in report.
- Demo games showing winning/defensive strategies.
- Analysis of pruning's impact on speed.

Project 4: Rule-Based Expert System for Crop Pest/Disease Diagnosis

Objective: Use knowledge representation and reasoning to build a practical tool for Kenyan smallholder farmers, applying logic to diagnose and advise on common agricultural issues.

Key Tasks:

- Create a knowledge base with 15–25 if-then rules for pests/diseases (e.g., fall armyworm on maize, banana wilt).
- Implement forward and/or backward chaining inference.
- Build interactive interface (console or simple input).
- Test with 5–10 example farmer scenarios.

Expected Outcomes:

- Accurate diagnosis and advice output.
- Report with rule examples and inference traces.
- Demo consultations.
- Discussion of limitations (e.g., incomplete knowledge in rural contexts).

Project 5: Sentiment Analysis on Kenyan Social Media Topics

Objective: Apply supervised machine learning to analyze public opinion in Kenya, evaluating models on local mixed-language data (English/Swahili/Sheng).

Key Tasks:

- Use/collect small dataset (~800–1500 samples) on a topic (e.g., M-Pesa fees, election views).
- Preprocess text (handle code-switching, stop words).
- Train/compare Naive Bayes and Logistic Regression.
- Compute accuracy, precision, recall, F1.

Expected Outcomes:

- Trained models with performance metrics table.
- Error analysis examples.
- Report on sentiment trends in Kenyan context.
- Ethical reflection on bias in local social media data.

Project 6: Crop Suitability or Quality Classifier Using Kenyan Data

Objective: Build and compare supervised classifiers on agriculture-related data, promoting precision farming insights relevant to Kenyan smallholders.

Key Tasks:

- Use public/open dataset (e.g., simplified soil/rainfall/crop yield data for maize/tea regions).
- Perform EDA with visualizations.
- Train/compare KNN, Decision Tree, SVM.
- Use cross-validation and tune basic hyperparameters.

Expected Outcomes:

- Plots (scatter, feature importance).
- Model comparison table (accuracy, etc.).
- Recommendation of best model for Kenyan farming.

- Report with insights on feature impact.

Project 7: Neural Network for Kenyan Crop Disease Image Classification

Objective: Introduce neural networks by classifying plant diseases common in Kenya, mirroring mobile tools used by farmers.

Key Tasks:

- Use subset of PlantVillage or similar dataset (focus on maize/tomato diseases, 4–6 classes, 5,000–10,000 images).
- Build simple feedforward NN or basic CNN.
- Train with backpropagation; plot loss/accuracy curves.
- Evaluate on test set; show sample predictions.

Expected Outcomes:

- Model achieving reasonable accuracy (>80% if possible).
- Training curves and confusion matrix.
- Demo of classifying uploaded images.
- Explanation of neural network basics in report.

Project 8: SMS Spam/Fraud Detector for Kenyan Mobile Money

Objective: Develop an ML-based text classifier to detect fraud in Kenyan SMS context (high M-Pesa usage), addressing cybersecurity in mobile finance.

Key Tasks:

- Use SMS Spam Collection + synthetic Kenyan examples (e.g., fake M-Shwari loans).
- Feature extraction (TF-IDF, bag-of-words).
- Train Naive Bayes/Logistic Regression.
- Evaluate metrics; test on mixed-language samples.

Expected Outcomes:

- Classifier with high F1-score on fraud detection.
- Examples of detected/undetected messages.
- Report on Kenyan scam patterns.
- Discussion of real-world challenges (e.g., false positives).

Project 9: AI Ethics Case Study on Kenyan Applications

Objective: Critically analyze ethical issues in Kenyan AI deployments, combining research with a small implementation to demonstrate bias awareness.

Key Tasks:

- Select topic (e.g., AI in digital lending like Tala, traffic surveillance).
- Research 3–5 sources on biases/impacts.
- Create synthetic dataset showing bias (e.g., loan approval skewed by location/gender).
- Train simple classifier; show unfair outcomes.
- Propose mitigations.

Expected Outcomes:

- Summary of issues in Kenyan context.
- Code demo illustrating bias.
- Mitigation strategies in report.
- Reflection on societal effects (e.g., digital divide).

Grading Rubric (Total: 50 marks per project)

- **Problem Understanding & Relevance** (10 marks) — Clear objective, strong Kenyan context link, ties to course topics.
- Data Collection (5 marks): Gathers, develops, prepares a new dataset or augments an existing dataset.
- **Implementation Quality & Correctness** (15 marks) — Clean, error-free code; proper use of AI techniques; runs correctly.
- **Report Results, Analysis & Evaluation** (10 marks) — Meaningful metrics/graphs; insightful comparison/discussion; limitations addressed.
- **Ethical & Societal Reflection** (5 marks) — Addresses biases, Kenyan impacts, fairness/privacy concerns.
- **Demo/Presentation & Teamwork** (5 marks) — Effective short demo; evidence of pair contribution; originality/creativity.